Use Case 1: Register / Login

Goal	Register / Login
Actor	Student
Main flow	1. Student opens the Web Based ML Model Evaluator. 2. Student selects **Register** to create an account or **Login** to access an existing account. 3. Student enters the required credentials. 4. System validates the submitted information. 5. System creates the account or authenticates the student. 6. System displays the student dashboard. 7. Student can proceed to dataset upload and model evaluation.
Alternate Flow	**Invalid Login Credentials** At Step 4, if the credentials are invalid: 4a. System displays an error message and asks the student to retry. 4b. Student re-enters the credentials. 4c. Flow backtracks to **Main Flow Step 3**.

Use Case 2: Upload Dataset

Goal	Upload Dataset
Actor	Student
Main flow	1. Student clicks **Choose File**. 2. File browser opens. 3. Student selects a valid CSV file, such as students.csv. 4. Student clicks **Upload**. 5. System validates the CSV format, file size, and structure. 6. System stores the dataset and calculates metadata such as rows and columns. 7. System displays upload confirmation with the dataset metadata. 8. Student proceeds to Preview Dataset.
Alternate Flow	**Invalid Dataset File** At Step 5, if the selected file does not satisfy the required CSV conditions: 5a. System displays a validation message and asks the student to select another file. 5b. Student selects a different CSV file. 5c. Flow backtracks to **Main Flow Step 3**.

Use Case 3: Preview Dataset

Goal	Preview Dataset
Actor	Student
Main flow	1. System retrieves the uploaded dataset. 2. System displays the first few rows in a table. 3. System displays column names and available dataset information. 4. Student reviews the sample data to confirm that the dataset was read correctly. 5. Student proceeds to Preprocess Data.
Alternate Flow	**Dataset Preview Not Suitable** At Step 4, if the student determines that the uploaded dataset is not suitable: 4a. Student chooses to upload another dataset. 4b. System returns the student to the upload interface. 4c. Flow backtracks to **Upload Dataset – Main Flow Step 3**.

Use Case 4: Preprocess Data

Goal	Preprocess Data
Actor	Student
Main flow	1. System displays preprocessing options for missing values, categorical encoding, and feature scaling. 2. Student reviews the dataset and selects the required preprocessing options. 3. Student clicks **Apply Preprocessing**. 4. System validates the selected options. 5. System applies missing-value handling, categorical encoding, and feature scaling. 6. System displays preprocessing completion confirmation. 7. System stores or caches the processed dataset for subsequent steps. 8. Student proceeds to Select Target Column.
Alternate Flow	**Invalid Preprocessing Configuration** At Step 4, if the selected preprocessing options are not compatible with the dataset: 4a. System displays a validation message and asks the student to modify the options. 4b. Student changes the preprocessing options. 4c. Flow backtracks to **Main Flow Step 2**.

Use Case 5: Select Target Column

Goal	Select Target Column
Actor	Student
Main flow	1. System displays the available columns from the preprocessed dataset. 2. System shows column names, data types, and sample values. 3. Student selects the target column, such as **Result** for classification or **Price** for regression. 4. Student clicks **Set as Target**. 5. System validates the selected column. 6. System designates the selected column as target **y** and the remaining suitable columns as features **X**. 7. System detects whether the problem is classification or regression. 8. System displays target-selection confirmation. 9. Student proceeds to Set Train-Test Split.
Alternate Flow	**Invalid Target Selection** At Step 5, if the selected target column is not valid: 5a. System displays a message asking the student to select another column. 5b. Student selects a different valid target column. 5c. Flow backtracks to **Main Flow Step 3**.

Use Case 6: Set Train-Test Split

Goal	Set Train-Test Split
Actor	Student
Main flow	1. System displays a train-test split slider or input field. 2. System shows the default **80% training / 20% testing** ratio. 3. Student reviews or adjusts the ratio, such as 70-30. 4. Student clicks **Apply Split**. 5. System validates the split configuration. 6. System partitions the dataset into training and testing data while keeping features and target values aligned. 7. System displays the number of training and testing rows. 8. Student proceeds to Select ML Algorithm.
Alternate Flow	**Invalid Train-Test Split** At Step 5, if the split ratio is invalid or produces an unsuitable partition: 5a. System displays a validation message and asks the student to adjust the ratio. 5b. Student changes the train-test split. 5c. Flow backtracks to **Main Flow Step 3**.

Use Case 7: Select ML Algorithm

Goal	Select ML Algorithm
Actor	Student
Main flow	1. System displays the supported machine learning algorithms. 2. Student reviews the available algorithms for the detected problem type. 3. Student selects an algorithm, such as Logistic Regression, Decision Tree, Random Forest, or KNN. 4. System records the selected algorithm. 5. System displays the selected model configuration. 6. Student proceeds to Train Model.
Alternate Flow	**Incompatible Algorithm** At Step 3, if the selected algorithm is not suitable for the detected problem type: 3a. System displays a compatibility message and asks the student to select another algorithm. 3b. Student selects a compatible algorithm. 3c. Flow backtracks to **Main Flow Step 3**.

Use Case 8: Train Model

Goal	Train Model
Actor	Student with ML Service
Main flow	1. Student reviews the dataset, target, split, and selected algorithm. 2. Student clicks **Train Model**. 3. System sends the training request and prepared training data to the **ML Service**. 4. ML Service initializes the selected machine learning algorithm. 5. ML Service trains the model using the training partition. 6. ML Service generates predictions using the test partition and calculates the required evaluation metrics. 7. System stores the trained model and associated results. 8. System displays a training-completion message to the student. 9. Student proceeds to Evaluate Model or Compare Models.
Alternate Flow	**Training Configuration Failure** At Step 4, if the ML Service cannot initialize the selected model because the configuration is unsuitable: 4a. System displays a message asking the student to change the model configuration. 4b. Student returns to algorithm selection and chooses another compatible algorithm. 4c. Flow backtracks to **Select ML Algorithm – Main Flow Step 3**.

Use Case 9: Evaluate Model

Goal	Evaluate Model
Actor	Student and Instructor
Main flow	1. System retrieves the evaluation results for a trained model. 2. System displays the appropriate metrics. 3. For classification, the system displays Accuracy, Precision, Recall, F1-score, and Confusion Matrix. 4. For regression, the system displays RMSE, MAE, and R^2 Score. 5. System generates relevant visualizations such as metric charts, confusion matrix, or residual plots. 6. Student reviews the evaluation results. 7. Instructor can view the available evaluation results for review.
Alternate Flow	**Evaluation Cannot Be Completed** At Step 1, if valid evaluation results are not available for the selected model: 1a. System displays a message asking the student to train a model first. 1b. Student returns to model training. 1c. Flow backtracks to **Train Model – Main Flow Step 2**.

Use Case 10: Compare Models

Goal	Compare Models
Actor	Student
Main flow	1. System displays trained models available for the current dataset. 2. Student selects at least two trained models. 3. Student clicks **Compare**. 4. System retrieves the evaluation results for the selected models. 5. System displays the model performances side by side using a metric table and comparison chart. 6. Student reviews the results and identifies the better-performing model. 7. Student can select a model for download.
Alternate Flow	**Models Not Compatible for Comparison** At Step 4, if the selected models do not have compatible evaluation configurations: 4a. System displays a message asking the student to choose compatible models. 4b. Student returns to the model-selection list and selects different models. 4c. Flow backtracks to **Main Flow Step 2**.

Use Case 11: Download Model / Report

Goal	Download Model / Report
Actor	Student
Main flow	1. System displays download options for the trained model and evaluation report. 2. Student selects **Download Trained Model** or **Download Evaluation Report**. 3. System retrieves the requested model artifact or generates the evaluation report. 4. System sends the file to the student's browser. 5. Student saves the downloaded file locally. 6. The report contains model information, dataset information, preprocessing, split ratio, evaluation metrics, and relevant visualizations.
Alternate Flow	**Requested File Not Available** At Step 3, if the requested model or report is not available: 3a. System displays a message asking the student to select another available result. 3b. Student returns to the evaluation results. 3c. Flow backtracks to **Evaluate Model – Main Flow Step 1**.

Use Case 12: Manage Users

Goal	Manage Users
Actor	Admin
Main flow	1. Admin logs in to the administration area. 2. System displays the user-management interface. 3. Admin views the registered users. 4. Admin selects a user account to manage. 5. Admin performs an allowed management action, such as viewing, updating, or deactivating the account. 6. System saves the updated user information. 7. System displays confirmation of the completed action.
Alternate Flow	**Invalid User Management Action** At Step 5, if the requested action is not permitted: 5a. System displays a permission message. 5b. Admin selects another permitted action or another user. 5c. Flow backtracks to **Main Flow Step 3**.

Use Case 13: Manage Datasets

Goal	Manage Datasets
Actor	Admin
Main flow	1. Admin opens the dataset-management area. 2. System displays stored dataset records and metadata. 3. Admin selects a dataset record. 4. Admin reviews or manages the dataset record according to administrative permissions. 5. System applies the requested management action. 6. System displays confirmation.
Alternate Flow	**Dataset Management Action Not Permitted** At Step 4, if the selected action is not permitted: 4a. System displays a permission message. 4b. Admin selects another dataset or permitted action. 4c. Flow backtracks to **Main Flow Step 3**.

Use Case 14: View System Logs

Goal	View System Logs
Actor	Admin
Main flow	1. Admin opens the system-log interface. 2. System retrieves available application activity logs. 3. System displays relevant events such as uploads, model training activity, and administrative actions. 4. Admin reviews the logs to monitor system activity. 5. Admin returns to the administration dashboard.
Alternate Flow	**Required Log Entry Not Found** At Step 2, if the requested activity is not available in the retrieved logs: 2a. System displays a message indicating that no matching log entry was found. 2b. Admin changes the search/filter criteria. 2c. Flow backtracks to **Main Flow Step 2**.